

**51B51/1.5M-E1-L-T-X-NEG**

Description : 5151 mm NaI(Tl) scintillation detector with integrated voltage divider, LED and temperature sensor.

Scintillation crystal	:	NaI(Tl), Ø51 x51 mm.
Photomultiplier tube	:	Ø38 mm, 10 stage Hamamatsu R10601-100
Drawing	:	VS-1465-90
Dimensions	Length	: 133 mm.
	Diameter	: 58 mm.
Electronics	:	Voltage divider (see below)
High voltage polarity	:	Negative
Electrical Connections	:	High voltage : 20 cm RG 174 marked RED Anode signal: 20 cm RG 174 marked YELLOW
Resistor chain	:	5.85 MΩ.
Temperature range	:	-20 °C to +65 °C.
Temperature sensor	:	DS18B20 Vdd brown Ground (black) DQ (white) 25 cm long silicon wires AWG30
LED	:	SSL LX 3044 USBC Cathode blue Anode Red 25 cm long silicon wires AWG30

Performance

Energy resolution	:	< 7.5% FWHM @ 662 keV.
Gain stability	:	< 1.0% over 24h at 20 °C
Typical high voltage	:	- 600 to 1000 V.
Energy range	:	20 keV – 3 MeV.

Last revision: 04-07-2024
Revision number:0
Document Name: 51B51_1.5M-E1-T-X-NEG

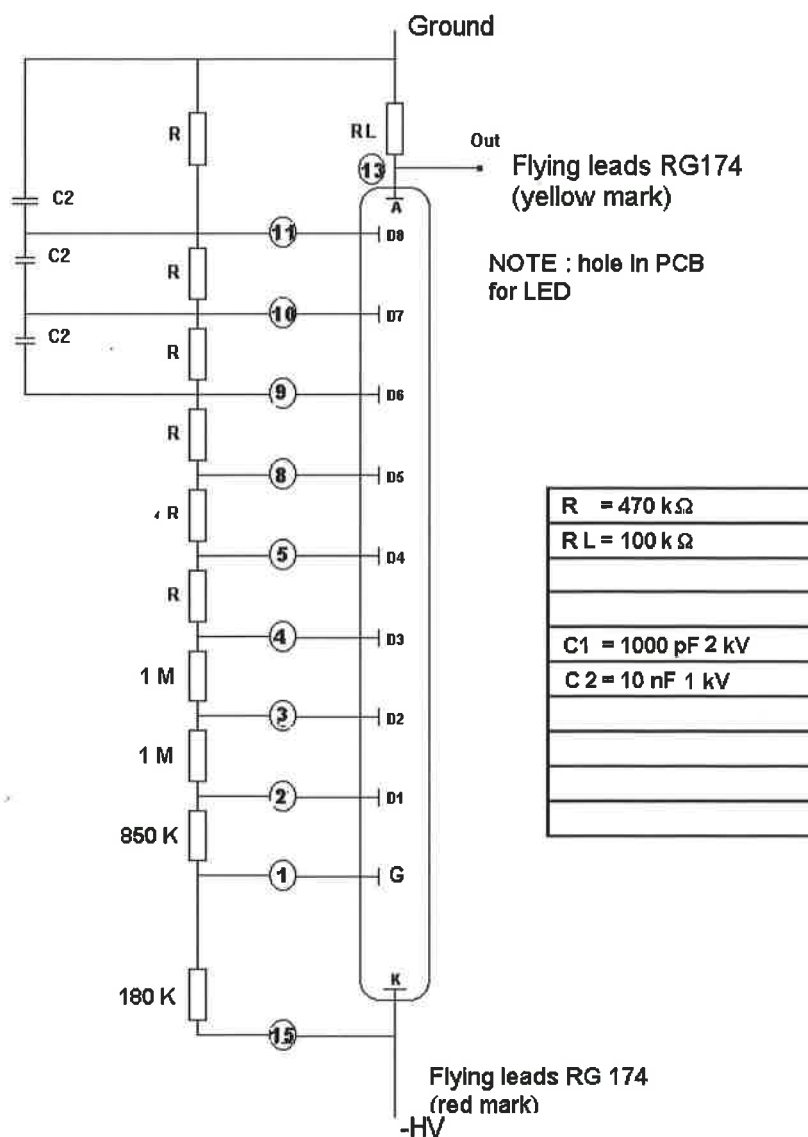
Scionix Holland B.V.

**Regulierenring 5
3981 LA Bunnik
The Netherlands**

**Tel: + 31 30 657 03 12
Website: www.scionix.nl
E-mail: sales@scionix.nl**

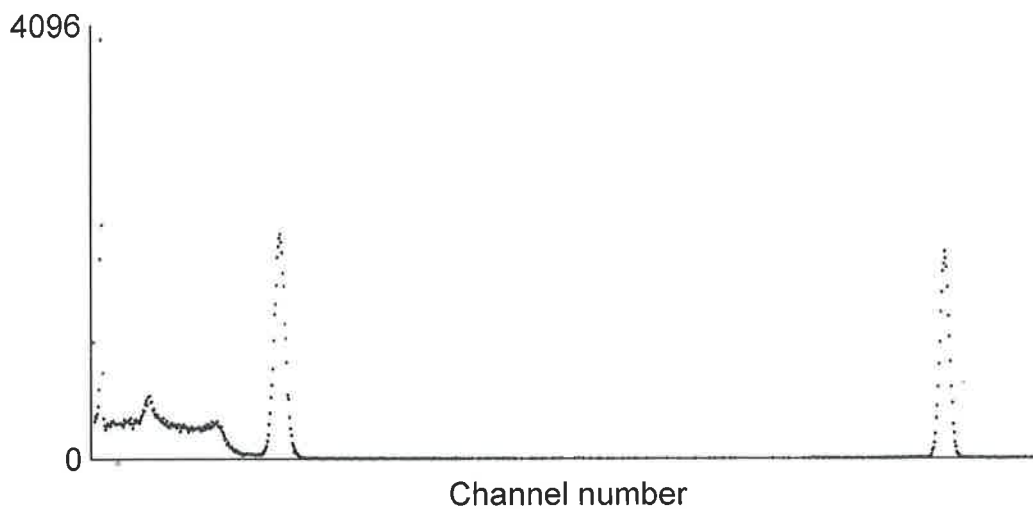
LED peak : A 3-3.5 V pulse amplitude, 200-250 ns long produces a reference Pulse at 2.5 - 3 MeV in the pulseheight spectrum

Voltage Divider



TEST SHEET

Detector type : 51B51/1.5M-E1-L-T-X-NEG
Crystal : NaI(Tl)
Read-out : Hamamatsu 1 1/2" Type R10601-100
Serial number : S1AB5195

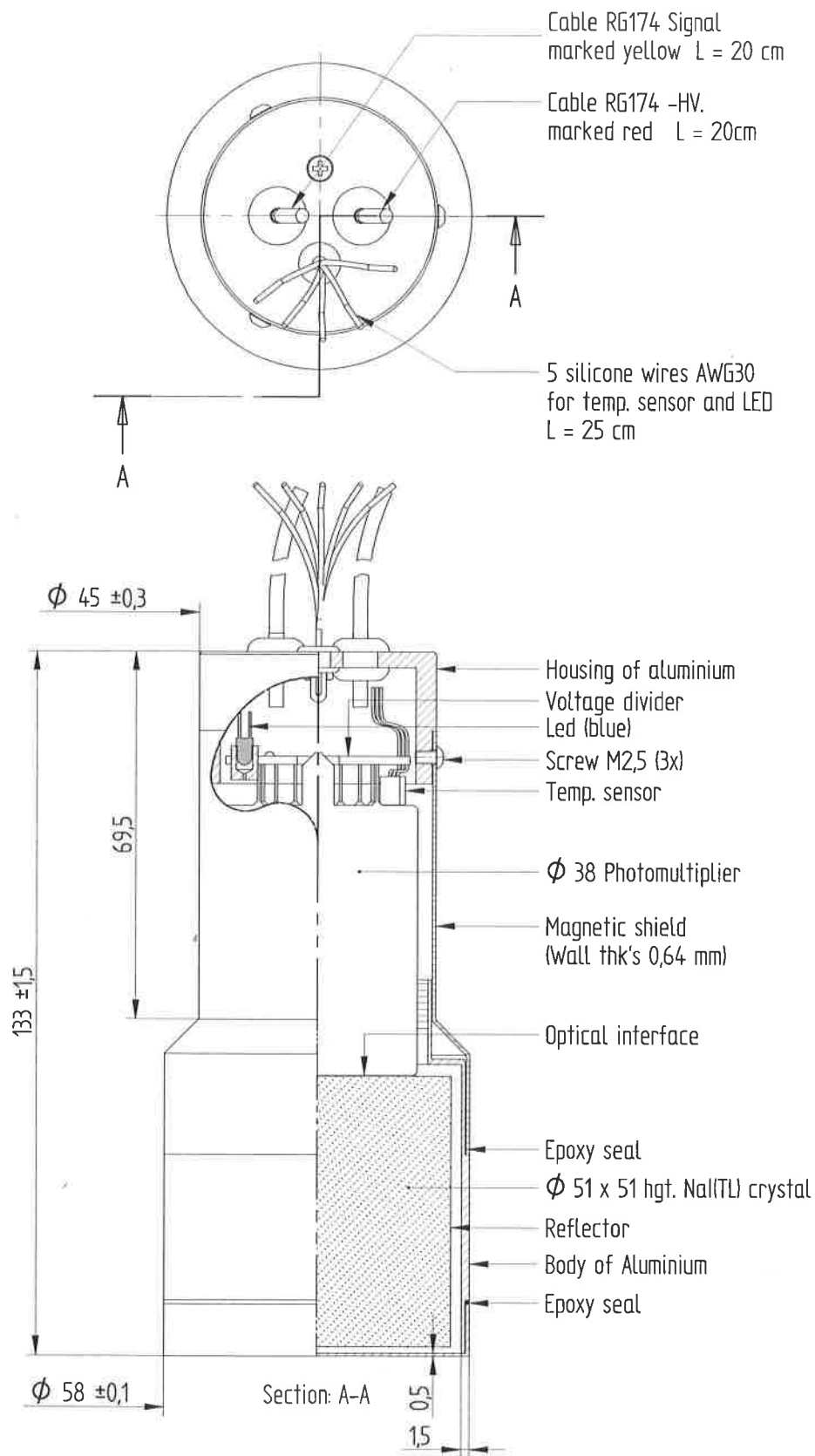


TEST RESULTS

Nuclide : Cs-137 (662 keV)
Energy resolution : 6.6 %
Noise : N/A
Peak to valley : N/A
GEE : 2.96 MeV
High voltage : -570 V
Date : 9 September 2024
Remarks : LED pulser: 200ns with 3.1V Amplitude
Tested by : Rien Parhan

Signature :

A handwritten signature in blue ink, consisting of a stylized 'R' followed by a series of loops and a horizontal line.



DRAWN: AS		DIMENSIONS: MM	REVISION	SPECIFICATION	DATE
SCALE: 1 : 1,2	DATE: 26-06-2024		R4		
Scionix Holland B.V. Regulierrening 5 3981 LA Bunnik The Netherlands Tel: +31 30 657 03 12 www.scionix.nl		 Dedicated Scintillation detectors		R3	
				R2	
				R1	
				TYPE: 51B51/1.5M-E1-L-T-X-NEG	
TITLE: ASSEMBLY		DWG. No. VS-1465-90			A4